

Clinical Risk Assessment Report: Patient 4

1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Positive for Metastasis
- Prediction Confidence: High Confidence
- Absolute Predicted Risk: 49.8% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 2.25x the diagnostic threshold

Clinical Takeaway:

The machine learning model classified this patient as High-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Large Zone Emphasis (CT) (GLZLM_LGZE.CT), which elevated the patient's absolute risk by +5.9%. Biologically, this feature reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

2. Key Pathological & Imaging Drivers (Elevating Risk)

Large Zone Emphasis (CT) (GLZLM_LGZE.CT) **Value: -0.39 [Bottom 5%] (+5.9%)**
Reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

Short Run High Gray-Level Emphasis (CT) (GLRLM_SRHGE.CT) **Value: 1.07 [Top 11%] (+4.8%)**
Reflects fragmented, short streaks of high density or metabolism, indicating a disjointed and aggressive tumor architecture.

Patient Age (Age) **Value: 48.0 [Bottom 7%] (+4.3%)**
Patient age influences immune surveillance, tissue microenvironment, and baseline metabolic rates.

Small Zone Emphasis (PET) (GLZLM_SZE.PET) **Value: 0.57 [Avg (73th %ile)] (+4.1%)**
Reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

Maximum Tumor Density (CONVENTIONAL_HUmax) **Value: 2.7 [Top 1%] (+2.8%)**
Reflects the most dense solid components or calcifications within the primary tumor lesion.

High Gray-Level Zone Emphasis (CT) (GLZLM_HGZE.CT) **Value: 1.17 [Top 9%] (+2.2%)**
Highlights the distribution of high-density or highly metabolic active regions clustered within the tumor.

75th Percentile Tumor Density (CONVENTIONAL_HUQ3) **Value: 0.69 [Avg (68th %ile)] (+2.2%)**
Reflects the overall physical density, solid components, and general attenuation characteristics of the tumor.

Squamous Cell Carcinoma Antigen (SCC_Ag) **Value: 0.7 [Avg (28th %ile)] (+2.1%)**
Serum levels reflect tumor proliferation, burden, and squamous cell differentiation.

3. Protective / Mitigating Factors (Reducing Risk)

No major risk-reducing features observed in the top drivers.

Machine Learning Feature Attribution (SHAP Decision Path)

